Clinical Intelligence Agent — Deployment Stack (ECS vs EKS)

Meeting note · 2026-09-14 · Recommendation: **ECS Fargate + S3/CloudFront**. EKS only if Kubernetes is mandated.

One-liner: Frontend and backend deploy and scale separately; UI on CDN/object storage, API as containers; files and secrets off the box; RDS later for doctor login.

1. What we deploy (same on both paths)

Piece	What	One-liner
Frontend	Doctor SPA	Static UI for conversation, SOAP, PDFs.
Backend	FastAPI + LangGraph + RAG	Runs the clinical pipeline and returns JSON + reports.
OpenRouter	deepseek/deepseek-chat	LLM for summary, entities, diagnosis, insights.
Deepgram	Nova ASR	Speech-to-text and speaker diarization.
LangSmith	Tracing	Observability for LangGraph and LLM calls.
Data	Audio, PDFs, FAISS, visits	Must live in S3/RDS in cloud, not local disk.

2. Shared AWS objects (both ECS and EKS)

Service / object	One-liner
Route 53	DNS for app.domain.com and api.domain.com.
ACM	Free TLS certificates for HTTPS.
S3 (frontend)	Stores the built UI files.
CloudFront	CDN in front of the UI; scales without servers.
S3 (data)	Audio, SOAP/analysis PDFs, guideline/FAISS artifacts.
ECR	Stores the backend Docker image by git SHA.
ALB	Public HTTPS entry for the API (and EKS Ingress).
VPC + security groups	ALB public; backend only reachable from the load balancer.
Secrets Manager	Holds Deepgram, OpenRouter, LangSmith, DB, JWT keys.
CloudWatch Logs	Collects container/pod logs.
GitHub Actions	Two pipelines: frontend ≠ backend deploys.
RDS Postgres (phase 2)	Doctor users, sessions, visit ownership, checkpoints.
AWS Budget	Monthly spend alert so LLM + infra do not surprise us.

3. ECS + Fargate — extra objects

One-liner: Cluster is a folder; Fargate runs tasks; the Service keeps N copies healthy behind the ALB.

Object	One-liner
ECS cluster	Logical group of services; almost no extra fee by itself.
Task definition	Recipe: CPU, memory, image, env, secrets for one backend container.
Fargate	AWS runs the container; we do not manage EC2 VMs.
ECS service	Keeps desired task count, rolling deploys, attaches to ALB.
Target group	ALB health-checks / or /health and sends traffic to healthy tasks.
Application Auto Scaling	Adds/removes tasks on CPU or ALB request count.
SQS + worker service (later)	Heavy analyze jobs off the API; workers can scale to 0.
NAT Gateway (optional)	Only if tasks sit in private subnets; adds cost.

4. EKS — extra objects

One-liner: Same ALB/S3/ECR/RDS, plus a Kubernetes control plane, nodes, and pod objects (Deployment, Service, Ingress, HPA).

Object	One-liner
EKS cluster	Managed Kubernetes control plane; billed even if idle (~$72/mo).
Node group / Karpenter	EC2 workers that actually run pods.
Namespace frontend / backend	Isolates UI and API in the cluster.
Deployment	Declares replica pods (like an ECS service).
Service (ClusterIP)	Internal stable network name for pods.
Ingress	Maps app.* / api.* to Services via AWS Load Balancer Controller.
AWS LB Controller	Turns Ingress into an ALB.
HPA	Autoscales pods on CPU/memory.
PDB	Keeps at least one pod up during deploys.
metrics-server	Feeds CPU stats to HPA.
Cluster Autoscaler / Karpenter	Adds/removes nodes when pods cannot schedule.
IRSA (pod IAM role)	Pods access S3/Secrets/RDS without embedding keys.
Helm / Argo CD (optional)	GitOps: separate charts for frontend vs backend.
NAT Gateway (typical)	Private nodes need outbound NAT; extra monthly cost.

5. Side-by-side

Need	ECS Fargate	EKS
Frontend	S3 + CloudFront	Same, or nginx Deployment
Backend compute	Fargate tasks	Pods on nodes
Scale frontend	CloudFront (automatic)	HPA on FE deploy
Scale backend	ECS service autoscaling	HPA + node autoscaler
Always-on fee	No control-plane fee	EKS ~$72/mo
Ops	Lower	Higher
Our pick	Yes for this project	If org requires K8s

6. Cost one-liners (infra USD/month, approximate)

Path	Staging	Light prod	One-liner
ECS Fargate	$45–90	$90–180	Pay for ALB + running tasks.
EKS	$150–230	$200–330	Control plane + nodes even when quiet.
APIs (both)	$10–50 demo	$70–230 clinic	OpenRouter + Deepgram + LangSmith often beat compute.

7. Honest gaps before cloud HA

No RDS login yet; MemorySaver is in-RAM (lost on restart / multi-task); local output/ must move to S3. Fix these before more than one replica.

8. 45-second script

We keep frontend and backend separate. On ECS: UI on S3/CloudFront, API as Fargate tasks behind an ALB, images in ECR, secrets in Secrets Manager, files in S3, later RDS. Two GitHub pipelines. On EKS we add a cluster, Deployments, Ingress, HPA, and node autoscaling, with extra cost and ops. Recommendation: ECS Fargate now; EKS only if we standardise on Kubernetes. Biggest bill risk is still LLM and Deepgram.

Estimates are planning figures, not an AWS quote. Set a Budget alert on day one.